

Introduction to Quarto for Reproducibility

SCO-SOC Virtual Meeting 2026

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Analysis and Data Tools for Science



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Preamble

Online workshops can be challenging

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Consider keeping your video on (if possible)

- Kids? Pets? Spouses? No problem
- But ultimately, you need to be comfortable! (and you absolutely have the right to privacy)

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Ask Questions!

- Group trouble-shooting is really valuable
- If you have a problem, others may also (or may have it in the future)

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Screen-sharing

- I may ask you to share your screen with the group (feel free to decline)
- For privacy, close your email etc. Or just share your RStudio window

Introductions

Me and my creatures



My garden



What about you?

- Name
- Background (Role, Area of study, etc.)
- Familiarity with R or Programming
- Creatures (furry, feathery, scaly, green or otherwise)?

Getting Started

Today we're learning to create static HTML reports from R code
(but can also create websites, pdfs, and presentations—like this one!)

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- Share your work
- Ensure reproducibility

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- Share your work
- Ensure reproducibility
- Be nice to your future self

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 - *What did I do again?*

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- Share your work
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 - *What did I do again?*
 - *What were the results?*
 - *Did I remember to remove partial observations?*

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Why?

- Keep track of your code *and* results
- Share your work
- Ensure reproducibility
- Be nice to your future self
 - *What did I do again?*
 - *What were the results?*
 - *Did I remember to remove partial observations?*

Okay, what kind of report?

For example...

```
1 ## Setup
2 This is my **great** study.... I used these packages:
3
4 ```{r}
5 library(tidyverse)
6 ```
7
8 ## Loading data
9 These are the datasets I used
10
11 ```{r}
12 my_data <- read_csv("https://raw.githubusercontent.com/steffilazerte/NRI_7350/main/data/chorus.csv")
13 my_data
14 ```
15
16 This is what it looks like
17
18 ```{r}
19 #| fig-width: 6
20 ggplot(data = my_data, aes(x = urbanization, y = songs)) +
21   geom_point()
22 ```
```

Becomes...

Setup

This is my **great** study.... I used these packages:

```
1 library(tidyverse)
```

```
— Attaching core tidyverse packages — tidyverse 2.0.0 —
✓ dplyr      1.2.1      ✓ readr      2.2.0
✓ forcats    1.0.1      ✓ stringr    1.6.0
✓ ggplot2    4.0.3      ✓ tibble     3.3.1
✓ lubridate  1.9.5      ✓ tidyr      1.3.2
✓ purrr      1.2.2
— Conflicts — tidyverse_conflicts() —
✗ dplyr::filter() masks stats::filter()
✗ dplyr::lag()     masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

Loading data

These are the datasets I used

```
1 my_data <- read_csv(
2   "https://raw.githubusercontent.com/steffilazerte/NRI_7350/main/data/chorus.csv"
3 )
4 my_data
```

```
# A tibble: 51 × 3
  urbanization songs calls
      <dbl> <dbl> <dbl>
1      0.794     0    136
```

For another example...

```
1  ### Visual of Thresholds Calculations
2
3  > - Pink ribbon = 99% Confidence interval of latitudes predicted from GAM
4  > - Black lines in the ribbon are the upper and lower limit, the middle
5  line is the predicted latitude (from GAM model)
6  > - Transparent blue rectangles indicate the date ranges used to establish
7  the latitudes just after and just before migration.
8  > - Blue horizontal lines represent the latitude threshold for spring
9  migration (begin/end)
10 > - Orange horizontal lines represent the latitude threshold for fall
11 migration (begin/end)
12
13 ```{r}
14 #| fig-asp: 1
15 #| fig-width: 15
16 wrap_plots(g) + plot_layout(guides = "collect", nrow = 1)
17 ```
```

(Plus a bunch of other options)

Becomes...

Setup

Load Data

Estimate Dates

Background

1. Defining the threshold latitude

2. Segmented Regression

Calculate Latitudinal Thresholds

Visual of Thresholds Calculations

Getting Dates from Thresholds

Plot Migration Dates

By year

Save Data

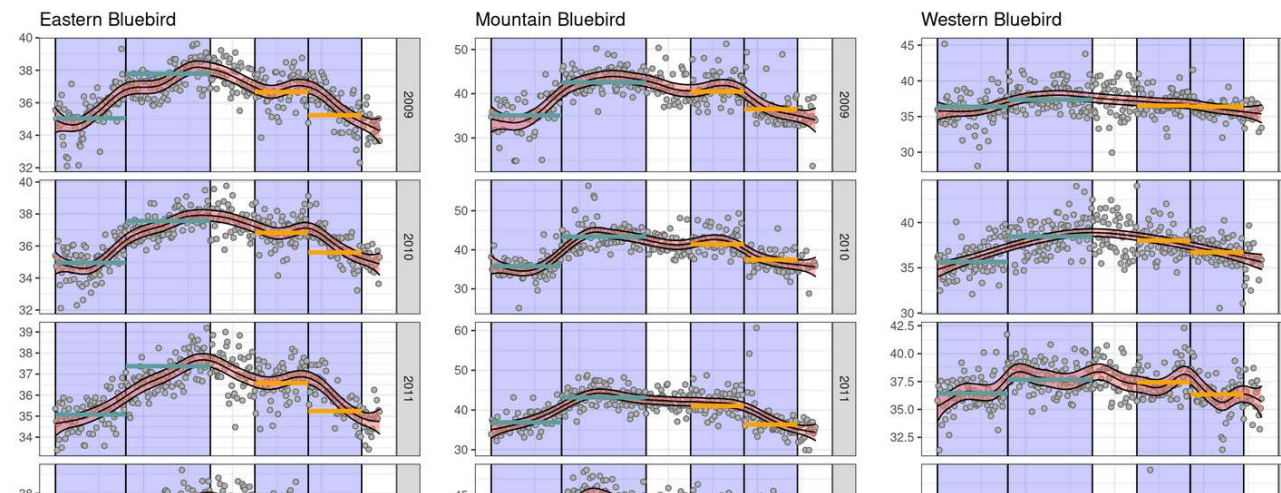
Reproducible

Session Info

Visual of Thresholds Calculations

- Pink ribbon = 99% Confidence interval of latitudes predicted from GAM
- Black lines in the ribbon are the upper and lower limit, the middle line is the predicted latitude (from GAM model)
- Transparent blue rectangles indicate the date ranges used to establish the latitudes just after and just before migration.
- Blue horizontal lines represents the latitude threshold for spring migration (begin/end)
- Orange horizontal lines represents the latitude threshold for fall migration (begin/end)

```
wrap_plots(g) + plot_layout(guides = "collect", nrow = 1)
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Wait a minute...

That doesn't look like an R Script...

Not an R script...

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Four things going on...

1. R code
2. R code fences (define code *chunks*)
3. Markdown
4. YAML *chunk* options

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Four things going on...

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This is actually not an .R script...
it's an R Markdown (**.Rmd**) or
Quarto (**.qmd**) document!

Quick start

- File > New Project > New/Existing Directory > Quarto Project
- Click “Render” button in the top panel (Quarto)
 - or “Knit” button (R Markdown)

Demo

Your Turn

Using this RStudio template, add in some code from your own scripts and render it.

Keep it relatively simple for now 😊

Your Turn

Using this RStudio template, add in some code from your own scripts and render it.

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What just happened? What are all these things?
R Markdown? Markdown? Quarto? YAML 😱

Terminology

R & RStudio

- Both are programs
- R is the programming language/environment
- RStudio is an IDE (integrated development environment) by Posit



R



RStudio

Terminology

R & Positron

- Both are programs
- R is the programming language/environment
- Positron is a *newer* IDE (integrated development environment) by Posit



R



Positron

Terminology

Markdown

- A text markup language
- Files are `.md`

For example, the following...

```
1  ### My heading
2
3  **Hi!** This is in italics
4
5  A [link](https://cran.r-project.org/) to R
```

Becomes...

My heading

Hi! This is in *italics*

A [link](https://cran.r-project.org/) to R

Terminology

R Markdown, Quarto, knitr, and Pandoc

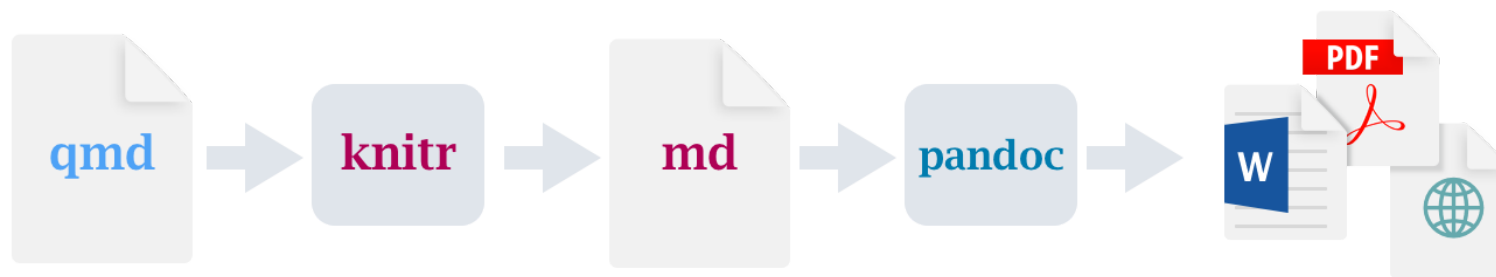
- R Markdown(`.Rmd`) and Quarto (`.qmd`) files are a *mix* of Markdown *and* R code
- **knitr** is an R package which evaluates R code and returns the output as a Markdown file
- **Pandoc** is a separate (independent) program that converts Markdown to a variety of formats
- **Quarto** manages the pipeline



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R Markdown vs. Quarto

Quarto (`.qmd`) is the *next generation* of R Markdown (`.Rmd`). You can still use R Markdown (it's not going anywhere), but Quarto is much newer and more powerful.

Terminology

YAML, HTML, CSS/SCSS

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- YAML is a language for specifying (meta)data
 - Used for specifying document options/configuration and *chunk* options

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 - Can be used directly in `.qmd/ .Rmd` files if you plan to output to HTML
 - E.g., can use `<br/>` for a line **break**

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YAML, HTML, CSS/SCSS

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 - Used for specifying document options/configuration and *chunk* options
- **HTML** is a language for making websites
 - Can be used directly in `.qmd/ .Rmd` files if you plan to output to HTML
 - E.g., can use `<br />` for a line **break**
- **CSS** is a language for *styling* websites
 - Can be used to apply custom styles to documents
 - **SCSS** is CSS with superpowers

Some options

Document level options - YAML block

```
1 ---
2 title: "My great analysis"
3 format: html
4 date: today
5 toc: true
6 code-fold: true
7 ---
```

- `date: today` to include today's date
- `toc: true` to include a table of contents
- `code-fold: true` to hide code (with option to show)

Note: These are Quarto options! R Markdown has similar ones, but they may be slightly different. E.g.,
`format: html_document` in R Markdown.

Some options

Chunk level options - YAML notation

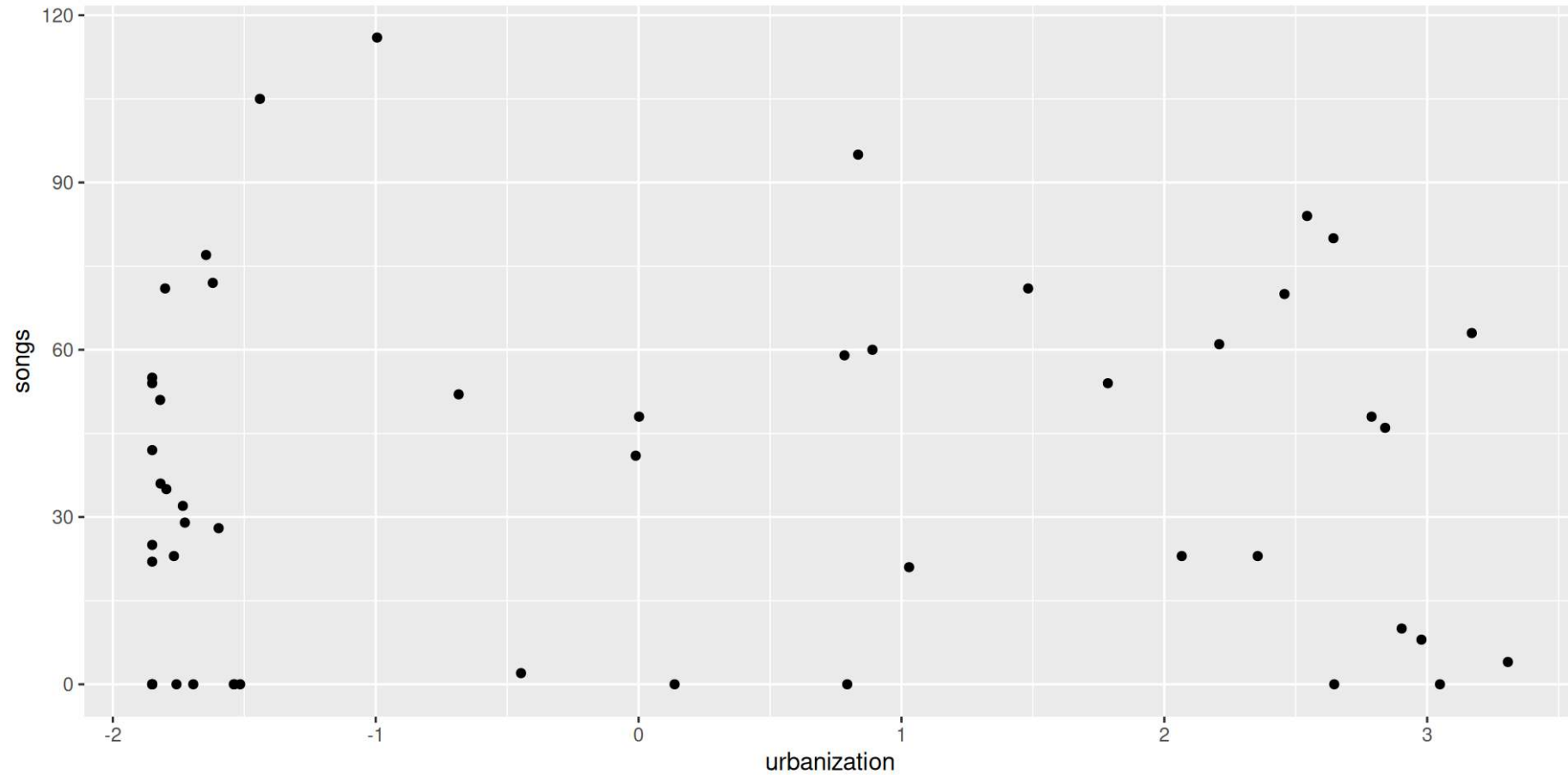
```
1  ```{r}
2  #| fig-width: 10
3  #| fig-asp: 0.5
4  #| fig-alt: |
5  #|   A scatterplot in black and white showing degree of
6  #|   urbanization on the x-axis and number of songs on
7  #|   the y-axis with no appreciable pattern in the data.
8  #| fig-cap: |
9  #|   The relationship between urbanization and the number
10 #|   of songs in mountain chickadee dawn choruses.
11
12 ggplot(data = my_data, aes(x = urbanization, y = songs)) +
13   geom_point()
14  ```
```

- `fig-width` width of figure in inches
- `fig-asp` aspect ratio of the figure (1 = square) (i.e. height = width * aspect ratio)
- `fig-alt` **Alternative** text for screen readers helping those who can't see the figure (should be descriptive, not the same as a caption)¹
- `fig-cap` Figure caption

1. More alt text resources: [when to include](#); [how to write](#); the [importance of context](#)

Gives...

```
1 ggplot(data = my_data, aes(x = urbanization, y = songs)) +  
2   geom_point()
```



The relationship between urbanization and the number of songs in mountain chickadee dawn choruses.

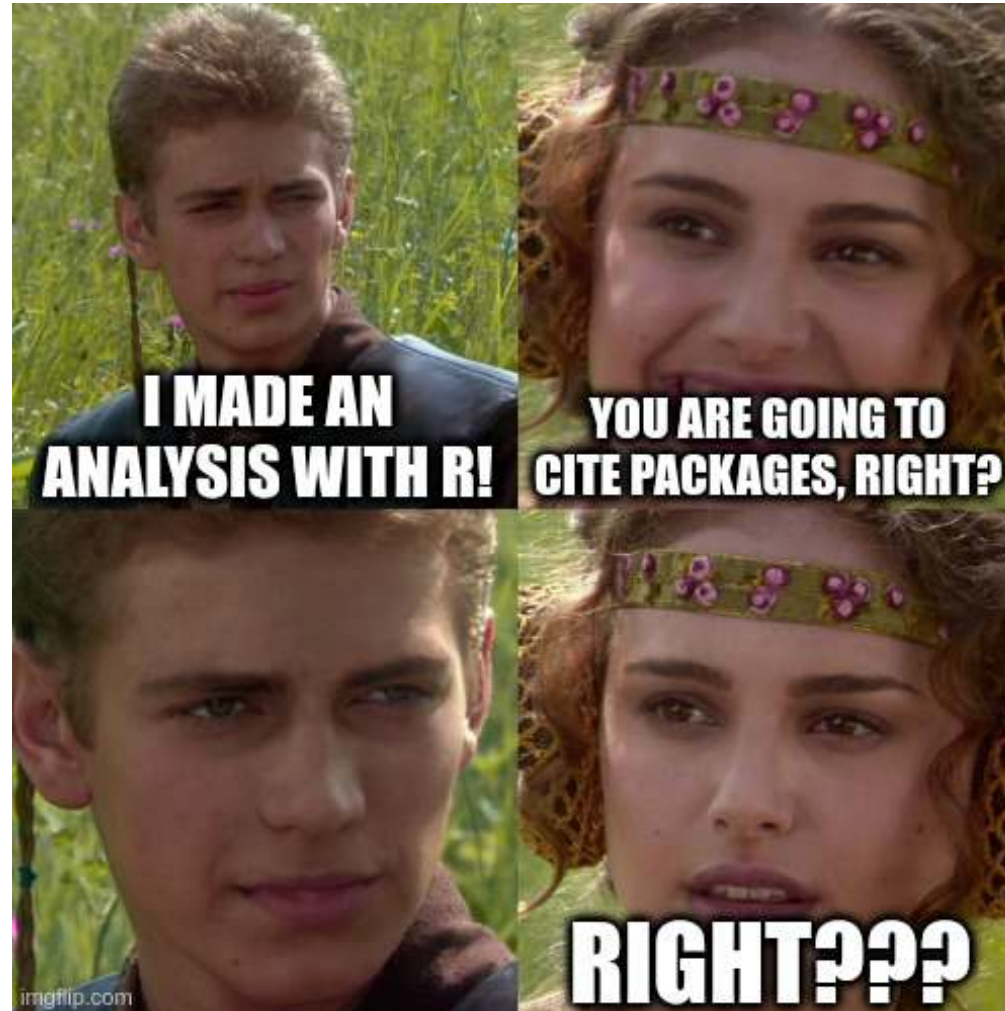
Enhancing reproducibility

- Create your publication figures through reports
- Date your reports (`my_analysis_2022-09-08.html`)
- Include info on packages used (because you're going to [cite](#) them... right? **RIGHT?**)
 - `devtools::session_info()`
 - `report::report_packages()`
 - `report::cite_packages()`
- Embed data directly (for smaller datasets) using [DT](#) package

```
1 DT::datatable(mtcars, extensions = 'Buttons',  
2               options = list(dom = 'Bfrtip', buttons = c('csv', 'excel')))
```

Cite the Packages!

Seriously, [cite](#) the packages 😁



Your Turn

Use the more advanced `template` (`example.qmd`) to create a reproducible report of your analysis.

Consider the options we learned

Anything you'd like to add?

Experiment with `code-fold: true` or `toc: false`

Some Final Thoughts

Rendering vs. Spinning

Rendering

`.Rmd/.qmd` → `.md` → HTML

Spinning

`.R` → `.md` → HTML

1. If you use R Markdown syntax instead of Quarto, you can use the “Compile Report” button

Rendering vs. Spinning

Rendering

`.Rmd/.qmd` → `.md` → HTML

- Easier to write narrative
- Define code
 - Use ````${r}```` and ````` to define code blocks
- Define options with `#|`
 - *Inside* a code block
- Render button or `quarto_render()` function

Spinning

`.R` → `.md` → HTML

- Easier to code
- Define text
 - Use `#'` to define markdown
- Define options with `#|`
 - Above a code block
 - Use `#| label: my-chunk` to split code chunks (just adds a label)
- `quarto_render()`¹ function

1. If you use R Markdown syntax instead of Quarto, you can use the “Compile Report” button

Rendering vs. Spinning

See `example_spin.R`

Rendering

```
1 ## Setup
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3
4 ```{r}
5 library(tidyverse)
6 ```
7
8 ## Loading data
9 These are the datasets I used
10
11 ```{r}
12 my_data <- read_csv("https://raw.githubusercontent.com/steffilaze
13 my_data
14 ```
15
16 This is what it looks like
17
18 ```{r}
19 #| fig-width: 6
20 ggplot(data = my_data, aes(x = urbanization, y = songs)) +
21   geom_point()
22 ```
```

Use “Render” button or render with:

```
1 quarto::quarto_render(
2   input = "example.qmd",
3   output_file = paste0("example_", Sys.Date(), ".html"))
```

Spinning

```
1 #' ## Setup
2 #' This is my **great** study.... I used these packages:
3
4 library(tidyverse)
5
6 #' ## Loading data
7 #' These are the datasets I used
8
9 my_data <- read_csv("https://raw.githubusercontent.com/steffilaze
10 my_data
11
12 #' This is what it looks like
13
14 #| fig-width: 6
15 ggplot(data = my_data, aes(x = urbanization, y = songs)) +
16   geom_point()
```

Render with:

```
1 quarto::quarto_render(
2   input = "example_spin.R",
3   output_file = paste0("example_spin_", Sys.Date(), ".html"))
```

Relative locations

If you use nested folders in your work, you'll want to use the [here](#) package to ensure all the file locations are consistent

```
1 library(here)
2 library(tidyverse)
3
4 my_data <- read_csv(here("Data/my_data.csv"))
```



Artwork by Allison Horst

Taking it to the next level

Analysis Websites

(Quarto Docs for Websites)

Vulture Migration ([GitHub](#) | [Website](#))

Vulture Migration on Vancouver Island

Workflow

Initial Exploration

Calculate Metrics

Analysis

Manuscript Figure and Supplemental

Appendices

Setup

Citations

Notes

Duration of migration

How long is migration? Has it changed in length?

- Look for changes in the number of days over which migration and peak migration occur
- `mig_dur_days`, `peak_dur_days`

Descriptive stats

Code

measure	mean	sd	min	median	max	n
mig_dur_days	24.29	3.43	16	25	30	24
peak_dur_days	9.54	1.32	6	10	11	24

Models

Figures

Model Checks

Sensitivity

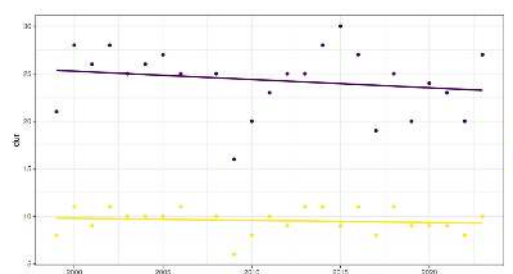
Problematic Years

Full Model Results

Note: Lines without Std Error ribbons are not significant

Code

`geom_smooth()` using formula = 'y ~ x'



On this page

Setup

Using This Report

Models

Model Results

Variables

Figures

DHARMA plots

Sensitivity

Timing of kettle formation

Duration of migration

Number of birds in kettles

Number of residents

Patterns of migration

Reproducibility

Edit this page

Report an issue

Urban Migration Stopovers ([GitHub](#) | [Website](#))

Code

```
walk(unique(trans$speciesID), \(x) {
  t1 <- filter(trans, speciesID == x)
  g <- ggplot(t1, aes(x = lon1, y = lat1)) +
    annotation_map_tile(type = "cartolight", zoomin = 0) +
    geom_spatial_segment(
      aes(xend = lon2, yend = lat2, colour = month),
      arrow = arrow(length = unit(10, "pt"), type = "closed",
        scale_colour_viridis_c(option = "inferno",
          begin = 0.20, end = 0.85) +
        guides(colour = guide_colourbar(reverse = TRUE)) +
        labs(title = t1$english[1],
          subtitle = paste0("n = ", n_distinct(t1$tagDeployID))
        print(g)
      })
})
```

Motus tracking of urban migration stopovers

Workflow

Selecting Projects

Download Data

Range Maps

Basic Filtering

Fine-scale Filtering

Bouts and Transitions

Finding Problems

Manual Adjustments

Summaries

Maps

Appendices

Notes & Details

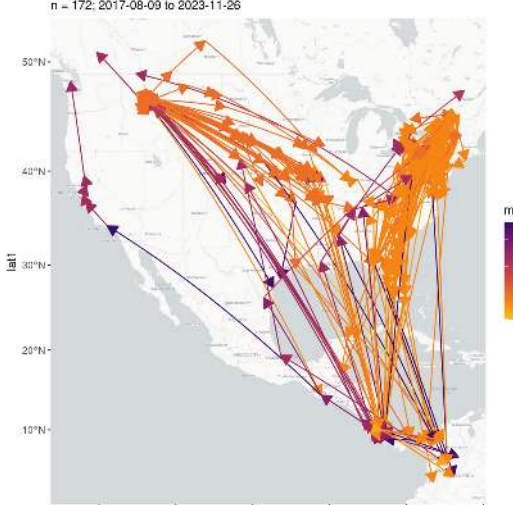
Setup

Datasets

Citations

Swainson's Thrush

n = 172: 2017-08-09 to 2023-11-26



On this page

Setup

Maps by species

Swainson's Thrush

Ovenbird

Hermit Thrush

Song Sparrow

Yellow-rumped Warbler (Myrtle)

Chestnut-collared Longspur

Purple Finch

Tennessee Warbler

Magnolia Warbler

Red-eyed Vireo

Sprague's Pipit

Western Meadowlark

Gray Catbird

White-throated Sparrow

Golden-crowned Sparrow

Blue-headed Vireo

Pine Siskin

Spotted Towhee

Northern Cardinal

Lazuli Bunting

Baird's Sparrow

Black-throated Blue Warbler

Common Yellowthroat

Northern Waterthrush

Horned Lark

Maps by individual

Swainson's Thrush

13854

13850

Manuscript Websites

(Quarto Docs for Manuscripts)

Effects of Riparian Grazing on Distinct Water-Extractable Phosphorus Sources ([GitHub](#) | [Website](#))

Table of contents

1 Introduction

2 Methods

3 Results and Discussion

3.1 Vertical profiles of P

3.2 Longitudinal profiles of P

3.3 Impacts of grazing and mowing on P sources

3.4 Sources of variability and uncertainty in P sources

3.5 Management implications

4 Conclusion

Acknowledgements

Data availability

Conflict of interest statement

Author contributions

References

Supplemental materials

Notebooks

Article Notebook

Correlation

Map of study area

Plant composition

Riparian litter WEP in response to grazing

Riparian organic and mineral soil WEP in response to grazing

Riparian vegetation WEP in response to grazing

Vertical profile of WEP

Weather

Workflow

3 Results and Discussion

3.1 Vertical profiles of P

The biomass, litter, organic layer, and Ah horizon sources of P demonstrated a strong vertical stratification in both the concentration and areal densities of WEP (Figure 2). The median concentrations in the vegetation sources were 82.8 and 39.0 $mg\ kg^{-1}$ for the biomass and litter components, respectively, which is more than an order of magnitude greater than the soil components (0.9 and 3.4 $mg\ kg^{-1}$; Ah and organic, respectively). Considerable variability in the WEP concentration in the biomass and litter sources were observed with interquartile ranges (IQR) of 54.3 and 32.9 $mg\ kg^{-1}$ for the biomass and litter sources, respectively. In contrast, the IQR for the organic and Ah sources both were $<2.5\ mg\ kg^{-1}$.

Overall, in terms of the areal density of WEP, the top 10 cm of the Ah horizon was the largest source of WEP ($42.5\ mg\ m^{-2}$) followed by the biomass ($26.3\ mg\ m^{-2}$), organic layer ($14.3\ mg\ m^{-2}$), and the litter ($13.7\ mg\ m^{-2}$). It should be noted that these are only approximate estimates for the organic layer and Ah horizon because the values for depth and bulk density measured in 2023 were used in the calculations for all previous years (Figure S4). Nevertheless, the vertical profile of WEP in riparian areas (Figure 2) observed in this study supports the concept that a measure of P in soil alone is likely missing a large proportion of the near-surface P that can be potentially lost during the spring snowmelt (Liu et al., 2019a; b; Cober et al., 2019). The substantial proportion of WEP above the soil surface provides evidence that managing the biomass in riparian areas in autumn may reduce the contribution of P lost directly from this area during spring. Specifically, the harvesting of this biomass results in an export of P which can maintain or enhance the buffering or storage capacity of P derived from upslope sources further improving downstream water quality (Kelly et al., 2007; Hille et al., 2019).

a)

b)

Figure 2: Vertical and longitudinal profiles of a) WEP concentration and b) WEP content in the riparian areas prior to grazing and mowing treatments.

The manuscript

Riparian vegetation WEP in response to grazing

In [17]:

```
ggplot(data = filter(plot_data, measure == "dryweight"), aes(x = treatment, y = diff, fill = location)) +  
  geom_boxplot() +  
  labs(y = "kg/m^2") +  
  geom_hline(yintercept = 0) +  
  facet_grid(~measure)
```

Table of contents

Load Libraries

Read in data

Merge data

Biomass data

Biomass Analysis

Dry Weight

Plots

dryweight

A notebook associated with the manuscript

34 / 37

What's next?

Quarto 2!

100x faster!

Collaborative editor!

Backwards compatibility for Quarto 1

Probably still about a year out

Thank You!

Resources

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